

Highlights

- Fast activated sludge surrogates can be accurate but physically inadmissible.
- Thirteen surrogates were tested across data scarcity and extrapolation regimes.
- Every raw surrogate set violated mass conservation; several predicted negatives.
- Projection made all 766,350 effluent predictions conserve mass and stay non-negative.
- Wastewater-process surrogate evaluation should separate error from feasibility.